

P 2.1

For each of the following systems, determine whether the system is (1) stable, (2) causal, (3) linear, (4) time invariant, and (5) memoryless:

(a) $T(x[n]) = g[n]x[n]$ with $g[n]$ given

(b) $T(x[n]) = \sum_{k=n_0}^n x[k] \quad n \neq 0$

(c) $T(x[n]) = \sum_{k=n-n_0}^{n+n_0} x[k]$

(d) $T(x[n]) = x[n - n_0]$

(e) $T(x[n]) = e^{x[n]}$

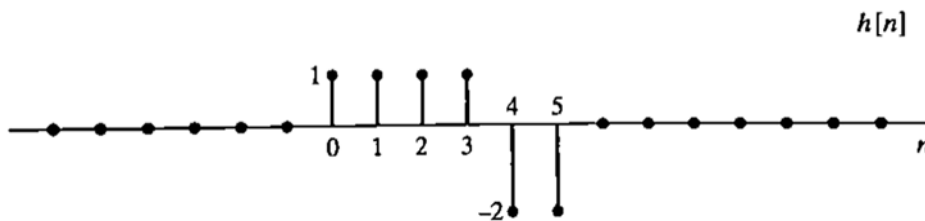
(f) $T(x[n]) = ax[n] + b$

(g) $T(x[n]) = x[-n]$

(h) $T(x[n]) = x[n] + 3u[n + 1]$.

P 2.26

The impulse response of a linear time-invariant system is shown in Figure P2.26. Determine and carefully sketch the response of this system to the input $x[n] = u[n - 4]$.


P 2.35

Consider an LTI system with frequency response

$$H(e^{j\omega}) = e^{-j(\omega - \frac{\pi}{4})} \left(\frac{1 + e^{-j2\omega} + 4e^{-j4\omega}}{1 + \frac{1}{2}e^{-j2\omega}} \right), \quad -\pi < \omega \leq \pi.$$

Determine the output $y[n]$ for all n if the input for all n is

$$x[n] = \cos\left(\frac{\pi n}{2}\right).$$

● Prove 2.9.4, 2.9.6 and 2.9.7

● Matlab: Find the solution of P2.25 (c) using Matlab and plot the result

P 2.25

For each of the pairs of sequences in Figure P2.25, use discrete convolution to find the response to the input $x[n]$ of the linear time-invariant system with impulse response $h[n]$.

